

GABRIELLE MAJETIC

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PROFILE SUMMARY

Motivated recent graduate in computational mathematics and statistics with 5 years of programming experience, 3 years of biomedical research experience, 2 years of experience with data science, and 1 year of experience with software engineering, database management, and machine learning. Quick, detail-oriented learner who can work independently.

SKILLS

Nucleic Acid Methods and Analysis

- DNA preparation from mice cells
- Size fractionation by agarose gel electrophoresis
- Gene specific polymerase chain reaction

Immunochemical Analysis

- Immunohistochemical methods

Programming/Software Methods

- Familiar languages: R, SQL, Java, Arduino, LabView, and Spike 2
- Advanced languages: Python, MATLAB/Octave, and NetLogo
- Familiar software: SAS, JMP, Microsoft Access
- Advanced software: Microsoft Office

Engineering Methods

- 3D modeling: Autodesk Fusion 360 and Inventor
- 3D printing: Preform software and Form 2 printer

Computational Concepts/Techniques

- Data science: data mining, statistical computing, data modeling, and data analysis
- Software engineering: GitHub/Git, unit testing, Scrum, debugging, and user documentation
- Database management: relational databases, relational algebra/calculus, SQL, entity-relationship models, and transaction control
- Machine learning: types of learning, components of a learning model, and learning algorithms

EDUCATION

Duquesne University **Pittsburgh, PA** **8/2021 – 5/2023**

- Degree: Master of Science in Computational Mathematics and Statistics
- Cumulative GPA: 3.93/4.00

Duquesne University **Pittsburgh, PA** **8/2016 – 5/2020**

- Degree: Bachelor of Science in Biomedical Engineering
- Minors: Mathematics and Biochemistry
- Certificate: Pre-Medical and Health Professions Programs
- Cumulative GPA: 3.89/4.00
- Honors: Magna Cum Laude

Thomas Jefferson High School **Jefferson Hills, PA** **8/2012 – 6/2016**

- Graduated in June 2016 with academic high honors

EMPLOYMENT

Duquesne University **8/2021 – 5/2023**

Graduate Assistant

- Supervisors: Alex Lipecky (8/2021 – 5/2022) and Larisa Shtrahman (8/2022 – 5/2023)
- Was a tutor for the following courses 8/2021 – 5/2022: Fundamentals of Statistics and Introduction to Biostatistics
- Was a teaching assistant for the following course 8/2022 – 12/2022: College Algebra
- Was a teaching assistant for the following courses 1/2023 – 5/2023: College Algebra/Trigonometry and Calculus I

Immunetrics

9/2020 – 8/2021

Part-Time Biologist Intern

- Supervisor: John Harn, QSP Modeling Platform Manager
- Found and captured applicable data in publications
- Converted applicable data to a format compatible with certain computational systems
- Organized applicable data such that future users can easily find, interpret the source of, and assess the level of trust to associate with that data

Duquesne University

5/2020 – 7/2020

Part-Time Laboratory Technician I

- Principal Investigators: Benedict Kolber, PhD, Biology and Rachael Miller Neilan, PhD, Mathematics
- Built a computational model using NetLogo coding software to simulate the behavior of cell-type specific and pain-related neurons in the entirety of the central nucleus of the amygdala (CeA)
- Conducted experiments utilizing the computational model and analyzed the resulting data
- Prepared a manuscript for publication of the resulting data from the computational model
- Provided statistical and programming support for other lab projects

YMCA of Greater Pittsburgh, Spencer Family Branch

5/2017 – 1/2020

Aquatics Swim Instructor: Privates and Group

5/2017 – 1/2020

- Supervisors: Morgan Clifford (5/2017 – 10/2019) and Stacey Lucidore (10/2019 – 1/2020)
- Taught children ages 6 mos. – 12 yr. with varying swimming abilities and/or learning disabilities how to become comfortable in the water, pool, beach, sun, and boat safety, rescue skills, and healthy life habits

Pool Day Camp Counselor

6/2017 – 8/2017

- Supervisor: Andrew Pricener
- Actively supervised children ages 5 yr. – 15 yr. during water activities and enforced camp and pool rules

YMCA of Greater Pittsburgh, South Hills/Cloverleaf Branch

6/2012 – 8/2016

Aquatics Swim Instructor: Group

- Supervisors: Joyce Card (6/2012 – 8/2015) and Andrew Pricener (5/2016 – 8/2016)
- Taught children ages 3 yr. – 12 yr. with varying swimming abilities and/or learning disabilities how to become comfortable in the water, pool, beach, sun, and boat safety, rescue skills, and healthy life habits

PAID RESEARCH EXPERIENCE

Duquesne University

5/2019 – 7/2019

2019 NIH NINDS Pain Undergraduate Research Experience (PURE) Fellow

- Principal Investigators: Benedict Kolber, PhD, Biology and Rachael Miller Neilan, PhD, Mathematics
- Built a computational model using NetLogo coding software to simulate the behavior of cell-type specific and pain-related neurons in the right CeA, and then used it to predict an individual's pain intensity as it evolves over time and in response to noxious stimulation
- Presented posters and oral presentations on this research

UNPAID (CREDITS TOWARDS DEGREE) RESEARCH EXPERIENCE

Duquesne University

1/2018 – 5/2020

Undergraduate Research Assistant

- Principal Investigators: Benedict Kolber, PhD, Biology and Rachael Miller Neilan, PhD, Mathematics
- Built a computational model using NetLogo coding software to simulate the behavior of cell-type specific and pain-related neurons in the entirety of the CeA, added features to test the impact of silencing certain neurons, added a neural network between neurons, and then used the model to predict an individual's pain intensity as it evolves over time and in response to noxious stimulation (Dr. Neilan and Dr. Kolber)
- Created an immunohistochemistry staining protocol for protein kinase C- δ to allow for cell-type identification of pain-related neurons in the CeA (Dr. Kolber)
- Designed and 3D printed containment units for Neurolux sensors to increase the number of times one Neurolux sensor can be used by an experimenter (Dr. Kolber)
- Optimized the operant pain tolerance assay experimental set-up for easier handling (Dr. Kolber)
- Genotyped transgenic mice to be used by other members in the lab for their experiments (Dr. Kolber)

HONORS AND AWARDS

- Duquesne University Undergraduate Research and Scholarship Virtual Competition 1st Place Office of Research Award for Outstanding Posters, **2020**
- National Institute of Mathematical and Biological Synthesis Undergraduate Research Conference Travel Award, **2019**
- Duquesne University Women in STEM Travel Award (\$500), **2019**
- Duquesne University PURE Directorship Award for Best Abstract, **2019**
- Duquesne University Dean's List for Biomedical Engineering, **2016 - 2020**

SELECT PUBLICATION

- Miller Neilan R, **Majetic G**, Gil-Silva M, Adke AP, Carrasquillo Y, Kolber BJ (2021) Agent-based modeling of the central amygdala and pain using cell-type specific physiological parameters. PLoS Comput Biol 17(6): e1009097. <https://doi.org/10.1371/journal.pcbi.1009097>

SELECT PRESENTATION

- **Majetic, G.**, Neilan, R., Adke, A., Carrasquillo, Y., Kolber, B. An Agent-Based Model of Pain-Related Neural Activity in the Amygdala. 2019 Biomedical Engineering Society Annual Meeting. Pennsylvania Convention Center, Philadelphia, PA. October 19, 2019 (poster)